10. Eigenvalues, and trace and determinant

Recall that for any 2×2 matrix $\mathbf{A},$ the characteristic polynomial can also be written in terms of the trace and determinant as follows:

$$\lambda^2 - (\mathrm{tr}\mathbf{A})\,\lambda + (\det\mathbf{A})\,.$$

Proof

For a matrix $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$ the characteristic polynomial is

$$\begin{aligned} \det(\lambda\mathbf{I} - \mathbf{A}) &= \det\begin{pmatrix} \lambda - a & -b \\ -c & \lambda - d \end{pmatrix} \\ &= (\lambda - a)(\lambda - d) - bc \\ &= \lambda^2 - (a+d)\,\lambda + (ad - bc) \\ &= \lambda^2 - (\mathrm{tr}\mathbf{A})\,\lambda + (\det\mathbf{A})\,. \end{aligned}$$

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For an $n \times n$ matrix $\mathbf{A},$ where $n \geq 2,$ it turns out that the characteristic polynomial, multiplied by ± 1 to make the leading coefficient 1, takes the form

$$\lambda^n - (\mathrm{tr}\mathbf{A})\,\lambda^{n-1} + \cdots \pm \det\mathbf{A}.$$

where the $\pm$ is $+$ if n is even, and $-$ if n is odd. So knowing $\mathrm{tr}\mathbf{A}$ and $\det\mathbf{A}$ determines 2 coefficients of the characteristic polynomial. More importantly, since

$$\lambda^n - (\mathrm{tr}\mathbf{A})\,\lambda^{n-1} + \cdots \pm \det\mathbf{A} = (\lambda - \lambda_1)(\lambda - \lambda_2)\cdots(\lambda - \lambda_n) \qquad (3.11)$$

where $\lambda_1, \cdots, \lambda_n$ are the n (not necessarily distinct) eigenvalues. Comparing coefficients gives

$$\mathrm{tr}\,(\mathbf{A}) = \lambda_1 + \lambda_2 + \cdots + \lambda_n \qquad (3.12)$$

$$\det\,(\mathbf{A}) = (\lambda_1)(\lambda_2)\cdots(\lambda_n). \qquad (3.13)$$

In other words, the trace is the **sum** of all n (not necessarily distinct) eigenvalues; the determinant is the **product** of all n eigenvalues.

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